

Prayas JEE 3.0 2025

Maths

DPP: 1

Trigonometric Functions

Q1 Find

$$\cos\left(\frac{\pi}{2} + \theta\right) + \cos\left(\frac{3\pi}{2} + \theta\right) + \cos\left(\frac{5\pi}{2} + \theta\right) \dots$$

$$+ \cos\left(\frac{7\pi}{2} + \theta\right) + \dots$$

upto 2021 terms.

- (A) $\sin \theta$
 (B) $\cos \theta$
 (C) $-\sin \theta$
 (D) $-\cos \theta$

Q2 Find value of

$$\sec(1140^\circ) + \operatorname{cosec}(1140^\circ) + \sin(1140^\circ) + \tan(1140^\circ) + \cot(1140^\circ) + \cos(1140^\circ)$$

- (A) $\frac{5}{2}(\sqrt{3} - 1)$
 (B) $\frac{5}{2}(\sqrt{3} + 1)$
 (C) $\frac{5}{2}(\sqrt{2} + 1)$
 (D) $\frac{5}{2}(\sqrt{2} - 1)$

Q3 The value of

$$\left(1 + \cos \frac{\pi}{6}\right) \left(1 + \cos \frac{\pi}{3}\right) \left(1 + \cos \frac{2\pi}{3}\right) \left(1 + \cos \frac{7\pi}{6}\right)$$

- is
 (A) $3/16$
 (B) $3/8$
 (C) $3/4$
 (D) $1/2$

Q4 If $\tan \alpha = \frac{m}{m+1}$ and $\tan \beta = \frac{1}{2m+1}$, then $\alpha + \beta =$

- (A) $\frac{\pi}{3}$
 (B) $\frac{\pi}{4}$
 (C) $\frac{\pi}{6}$

(D) None of these

Q5 If $\tan \alpha$ equals the integral solution of the inequality $4x^2 - 16x + 15 < 0$ and $\cos \beta$ equals to the slope of the bisector of first quadrant, then $\sin(\alpha + \beta) \sin(\alpha - \beta)$ is equal to

- (A) $\frac{3}{5}$
 (B) $-\frac{3}{5}$
 (C) $\frac{2}{\sqrt{5}}$
 (D) $\frac{4}{5}$

Q6 The value of $\cos 10^\circ - \sin 10^\circ$ is

- (A) Positive (B) Negative
 (C) 0 (D) 1

Q7 If

$$\sin x_1 + \sin x_2 + \sin x_3 + \dots + \sin x_{2021} = 2021$$

Then value of

$$\cos x_1 + \cos x_2 + \dots + \cos x_{2021} =$$

- (A) 1
 (B) 0
 (C) $\frac{\pi}{1}$
 (D) $\frac{\pi}{2}$

Q8 If $\sin A = \frac{3}{5}$ and $\cos B = \frac{-12}{13}$, where A and B both lie in second quadrant, then find the value of $\sin(A + B)$

- (A) $\frac{65}{56}$
 (B) $-\frac{65}{56}$
 (C) $-\frac{56}{65}$
 (D) $\frac{56}{65}$



- Q9** If $\tan \alpha + \cot \alpha = a$, then the value of $\tan^4 \alpha + \cot^4 \alpha$ is equal to
- (A) $a^4 + 4a^2 + 2$
(B) $a^4 - 4a^2 + 2$
(C) $a^4 - 4a^2 - 2$
(D) $-a^4 + 2a^2 + 4$
- Q10** If $\cos \theta + \sec \theta = -2$, $\theta \in [0, 2\pi]$, then $\sin^8 \theta + \cos^8 \theta$ equal to
- (A) -2 (B) 1
(C) 2^4 (D) 2^5



Answer Key

Q1 (C)

Q2 (B)

Q3 (A)

Q4 (B)

Q5 (D)

Q6 (A)

Q7 (B)

Q8 (C)

Q9 (B)

Q10 (B)



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